IB DIPLOMA PROGRAMME · 2024–2026

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COMPLETE REVISION GUIDE

IB Math AI Cheat Sheet

Full Syllabus · SL + HL · Score a 7

📐 Number & Algebra 📈 Functions 📏 Geometry & Trig 📊 Statistics 🎲 Probability

∫ Calculus

📖 How to Use This Sheet

NAVIGATION

- Each **section** matches an IB topic — use the colour-coded tabs
- **Formula boxes** (blue) = must-memorise formulas
- **Worked examples** (light blue) = follow step-by-step
- 💡 EXAM TIP = high-yield exam advice
- ⚠ **Warnings** = common errors to avoid
- 🔢 **Calc Tip** = TI-84 / GDC shortcuts
- HL = Higher Level only content

REVISION STRATEGY

- **Week before:** Read full sheet, highlight gaps
- **3 days before:** Focus on worked examples & tips
- **Night before:** Use Pages 19–20 only
- **Exam day:** Check calculator mode (degrees!)
- Prioritise topics by mark weight in your paper
- Always show method — partial marks exist

FULL SYLLABUS MAP

📐 Number & Algebra 1

Percentages & ratios
Sequences (AP, GP)
Exponents & logarithms
Financial mathematics
Amortization HL

📈 Functions 2

Linear & quadratic
Exponential & logarithmic
Graph transformations
Modelling & regression
Sinusoidal HL

📏 Geometry & Trig 3

Trigonometry (sin/cos/tan)
Sine & cosine rules
Area & volume formulas
Right-angle problems
Vectors HL

📊 Statistics 4

Descriptive stats
Normal distribution
Correlation & regression
Hypothesis testing
Chi-squared test

🎲 Probability 5

Basic rules
Conditional probability
Tree diagrams / Venn
Binomial distribution
Poisson HL

∫ Calculus 6

Differentiation basics
Optimisation
Integration
Trapezoidal rule
Differential equations HL

💡 **EXAM INSIGHT**

Paper 1 is non-calculator. Paper 2 is calculator. Paper 3 (HL only) is an extended problem-solving paper. Most real-world application questions appear in Paper 2.

Number & Algebra

PERCENTAGES & RATIOS

PERCENTAGE CHANGE

$$\% \text{ change} = (\text{new} - \text{old}) / \text{old} \times 100$$

Positive = increase, Negative = decrease

REVERSE PERCENTAGE

$$\text{Original} = \text{Amount} \div (1 \pm \%/100)$$

e.g. price after 20% tax: divide by 1.2

EXPONENTS & LOGARITHMS

LAWS OF EXPONENTS

$a^m \times a^n = a^{m+n}$ $\quad a^m \div a^n = a^{m-n}$

$(a^m)^n = a^{mn}$ $\quad a^0 = 1$

$a^{-n} = 1/a^n$ $\quad a^{(1/n)} = \sqrt[n]{a}$

LOGARITHM RULES

$$\log(AB) = \log A + \log B$$

$$\log(A/B) = \log A - \log B$$

$$\log(A^n) = n \cdot \log A$$

Change of base: $\log_b(x) = \ln(x)/\ln(b)$

SEQUENCES

ARITHMETIC PROGRESSION (AP)

$$u_n = u_1 + (n-1)d$$

$$S_n = n/2 \cdot (2u_1 + (n-1)d)$$

$$S_n = n/2 \cdot (u_1 + u_n)$$

d = common difference. S_n = sum of first n terms.

GEOMETRIC PROGRESSION (GP)

$$u_n = u_1 \cdot r^{n-1}$$

$$S_n = u_1(r^n - 1)/(r - 1), \; r \neq 1$$

$$S_\infty = u_1/(1 - r), \; |r| < 1$$

r = common ratio. S_∞ = infinite sum (converges).

✎ WORKED EXAMPLE — SEQUENCES

An AP has $u_1 = 3$, $d = 5$. Find u_{10} and S_{10}.

1. Identify: $u_1 = 3$, $d = 5$, $n = 10$
2. $u_{10} = 3 + (10-1)\times 5 = 3 + 45 = 48$
3. $S_{10} = 10/2 \times (3 + 48) = 5 \times 51 = 255$

⚠ COMMON MISTAKE

In GP, if asked for the *sum*, never use u_n formula. And check if $|r| < 1$ before using S_∞.

🔢 CALC TIP — GDC

Use **seq(** function on TI-84: `seq(3+5(X-1), X, 1, 10)` lists all terms.

💰 Financial Mathematics

INTEREST

SIMPLE INTEREST

$$I = P \times r \times t$$

$$A = P(1 + rt)$$

P = principal, r = annual rate (decimal), t = time (years)

COMPOUND INTEREST

$$A = P(1 + r/n)^{(nt)}$$

n = compoundings per year. Monthly: n=12, quarterly: n=4

CONTINUOUS COMPOUNDING

$$A = Pe^{(rt)}$$

e ≈ 2.718. Used in growth/decay models.

DEPRECIATION & GROWTH

DEPRECIATION

$$V = P(1 - r)^t$$

r = depreciation rate per period. This is a GP with ratio (1-r)

INFLATION / REAL VALUE

$$\text{Real value} = \text{Nominal} / (1 + i)^t$$

i = inflation rate. Buying power decreases over time.

AMORTIZATION (LOAN REPAYMENT) HL

$$R = PV \cdot i / (1 - (1+i)^{-n})$$

R = regular payment, PV = loan amount, i = rate per period, n = periods

GDC FINANCIAL APP

🔢 **TVM SOLVER (FINANCE APP)**

Press **APPS → Finance → TVM Solver**. Enter N (total periods), I% (annual rate), PV (present value, negative = outflow), PMT, FV, P/Y (payments/year), C/Y (compoundings/year). Solve for unknown by pressing ALPHA + ENTER on that field.

✎ WORKED EXAMPLE — COMPOUND INTEREST

€5000 invested at 4% p.a. compounded monthly for 6 years. Find final amount.

1. P = 5000, r = 0.04, n = 12, t = 6
2. `A = 5000 × (1 + 0.04/12)^(12×6)`
3. `A = 5000 × (1.003333...)^72 = 5000 × 1.2704 ≈ €6352`
4. GDC: TVM N=72, I%=4, PV=−5000, PMT=0, FV=? P/Y=12, C/Y=12 → FV = 6352.45

💡 **EXAM INSIGHT**

Always specify the compounding period. "4% per annum compounded monthly" means r/n = 0.04/12 per month.

⚠ **COMMON MISTAKE**

Don't confuse simple and compound interest. Simple: linear growth. Compound: exponential growth.

Functions

LINEAR

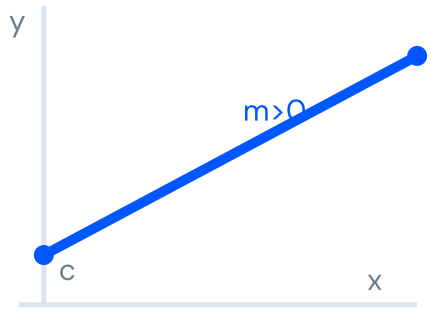
EQUATION

$$y = mx + c$$

m = slope = Δy/Δx
c = y-intercept

TWO POINTS

$$m = (y_2 - y_1)/(x_2 - x_1)$$



QUADRATIC

STANDARD FORM

$$y = ax^2 + bx + c$$

a>0: opens up, a<0: opens down

VERTEX FORM

$$y = a(x - h)^2 + k$$

Vertex at (h, k)

QUADRATIC FORMULA

$$x = (-b \pm \sqrt{(b^2 - 4ac)}) / 2a$$

Discriminant Δ = b²−4ac
Δ>0: 2 roots, Δ=0: 1, Δ<0: none

EXPONENTIAL

EXPONENTIAL FUNCTION

$$y = a \cdot b^x + c$$

b>1: growth, 0<b<1: decay
a = initial value (x=0 → y=a+c)

NATURAL EXPONENTIAL

$$y = Ae^{k\,x}$$

k>0: growth, k<0: decay
Half-life: t½ = ln2/k

LOGARITHMIC

$$y = a \cdot \ln(x) + b$$

Inverse of exponential. Domain x>0

KEY FEATURES TO IDENTIFY

Feature	How to Find	What it Means
x-intercepts (roots)	Set y = 0, solve for x	Where graph crosses x-axis
y-intercept	Set x = 0	Starting value / initial condition
Vertex / Turning point	x = -b/2a (quadratic) or GDC	Max or min of function
Asymptote	Exponential: y = c; Logarithmic: x = 0	Line function approaches but never crosses
Domain & Range	Read from graph or algebra	Valid inputs & possible outputs

🔢 **GDC FUNCTION TIPS**

Roots: 2nd → CALC → 2:zero. **Max/Min:** 2nd → CALC → 3:min or 4:max. **Intersection:** 2nd → CALC → 5:intersect. Always set window first: WINDOW key.

Transformations & Modelling

GRAPH TRANSFORMATIONS

Transformation	Effect on f(x)	Example
Vertical translation	$f(x) + k$	Shifts up k units (k>0)
Horizontal translation	$f(x - h)$	Shifts right h units (h>0)
Vertical stretch	$a \cdot f(x)$	a>1 stretches, 0<a<1 compresses
Horizontal stretch	$f(bx)$	b>1 compresses, 0<b<1 stretches
Reflection x-axis	$-f(x)$	Flips vertically
Reflection y-axis	$f(-x)$	Flips horizontally

SINUSOIDAL FUNCTIONS HL

SINUSOIDAL MODEL

$$y = a \cdot \sin(b(x - c)) + d$$

a = amplitude = (max−min)/2
b = 2π/period → period = 2π/b
c = horizontal shift (phase)
d = vertical shift = (max+min)/2

FINDING SINUSOIDAL PARAMETERS

1. Find max and min values from data/graph
2. $a = (max-min)/2$, $d = (max+min)/2$
3. Find period T, then $b = 2\pi/T$
4. Find c from when function first reaches maximum

MODELLING WITH FUNCTIONS

Choosing the Right Model

- **Linear:** constant rate of change, straight graph
- **Quadratic:** parabola, projectile motion, area problems
- **Exponential growth:** population, bacteria, compound interest
- **Exponential decay:** radioactive decay, depreciation
- **Logarithmic:** diminishing returns, pH, Richter scale
- **Sinusoidal:** temperature, tides, oscillations

EXAM INSIGHT

Modelling questions ask you to: (1) state the model, (2) find parameters, (3) interpolate/extrapolate, (4) comment on validity/limitations.

COMMON MISTAKE — TRANSFORMATIONS

In f(x−h), the shift is RIGHTWARD when h is positive. Students often shift the wrong direction. Remember: f(x−3) shifts 3 units RIGHT, not left.

Geometry & Trigonometry

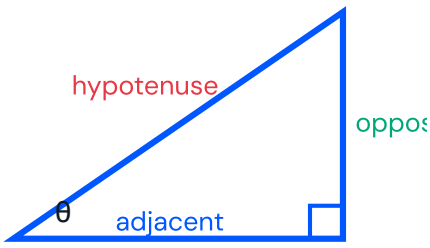
RIGHT-ANGLE TRIGONOMETRY

SOH CAH TOA

$\sin\theta = \text{opp}/\text{hyp}$

$\cos\theta = \text{adj}/\text{hyp}$

$\tan\theta = \text{opp}/\text{adj}$



PYTHAGOREAN THEOREM

$a^2 + b^2 = c^2$

c = hypotenuse (opposite right angle)

NON-RIGHT TRIANGLE RULES

SINE RULE

$a/\sin A = b/\sin B = c/\sin C$

Use when: AAS, ASA, or SSA (ambiguous case!)

COSINE RULE

$a^2 = b^2+c^2 - 2bc\cdot\cos A$

$\cos A = (b^2+c^2-a^2) / 2bc$

Use when: SAS, SSS

AREA OF TRIANGLE

$\text{Area} = \frac{1}{2}\cdot a\cdot b\cdot \sin C$

C = included angle between sides a and b

WORKED — SINE RULE

Find side a, given A=35°, B=65°, b=12 cm

1. Apply sine rule: $a/\sin 35° = 12/\sin 65°$
2. $a = 12 \times \sin 35°/\sin 65° = 12 \times 0.5736/0.9063 \approx 7.59$ cm

AREA & VOLUME FORMULAS

Shape	Area	Volume	Note
Rectangle	$l \times w$	$l \times w \times h$	Cuboid
Triangle	$\frac{1}{2} \times b \times h$	—	h = perpendicular height
Circle / Cylinder	πr^2	$\pi r^2 h$	Curved SA: $2\pi rh$
Sphere	$4\pi r^2$	$4/3\ \pi r^3$	From formula booklet
Cone	$\pi rl + \pi r^2$	$1/3\ \pi r^2 h$	l = slant height = $\sqrt{(r^2+h^2)}$
Pyramid	Sum of faces	$1/3 \times$ base area $\times h$	Any base shape
Trapezoid	$\frac{1}{2}(a+b)h$	—	a, b = parallel sides

Geometry: Applications & 3D Problems

BEARINGS & ANGLES OF ELEVATION

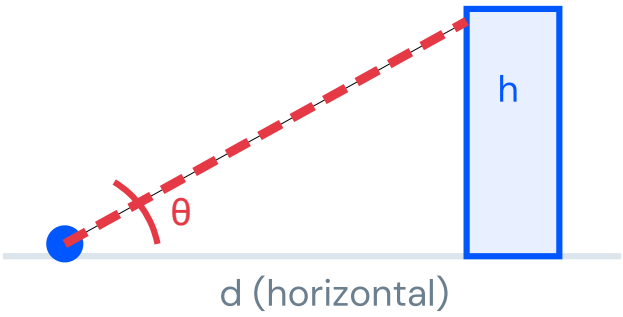
Bearing Rules

- Bearings measured **clockwise from North**
- Always written as 3-digit angle: 005°, 090°, 270°
- Back-bearing = bearing ± 180°
- Draw a diagram first — always!

ANGLE OF ELEVATION / DEPRESSION

$$\tan\theta = \frac{\text{height}}{\text{horizontal distance}}$$

Elevation: looking up. Depression: looking down.



3D GEOMETRY

3D Diagonal (Cuboid)

$$d = \sqrt{l^2 + w^2 + h^2}$$

Angle in 3D

- Identify the right triangle in 3D figure
- Find the base (horizontal distance) first
- Then use trigonometry on the vertical triangle
- Always draw 2D cross-sections separately

VECTORS HL

VECTOR OPERATIONS

$$|\mathbf{v}| = \sqrt{v_1^2 + v_2^2 + v_3^2}$$

$$\mathbf{v}\cdot\mathbf{w} = v_1w_1 + v_2w_2 + v_3w_3$$

$$\cos\theta = \frac{\mathbf{v}\cdot\mathbf{w}}{|\mathbf{v}||\mathbf{w}|}$$

Dot product = 0 → vectors are perpendicular

VECTOR EQUATION OF LINE

$$\mathbf{r} = \mathbf{a} + t\mathbf{b}$$

a = point on line, b = direction vector, $t \in \mathbb{R}$

WORKED EXAMPLE — 3D PROBLEM

A cuboid has l=8, w=6, h=4. Find the angle the space diagonal makes with the base.

1. Base diagonal: $d_{base} = \sqrt{8^2+6^2} = \sqrt{100} = 10$
2. Space diagonal: $d_{space} = \sqrt{10^2+4^2} = \sqrt{116} \approx 10.77$
3. Angle with base: $\theta = \arctan(4/10) = 21.8°$

Statistics: Descriptive Measures

MEASURES OF CENTRAL TENDENCY

MEAN ($\bar{x}$)

$$\bar{x} = \Sigma x / n = \Sigma(f \cdot x) / \Sigma f$$

Affected by outliers. Best for symmetric data.

MEDIAN

Middle value when data is ordered.

n values: position = (n+1)/2

Not affected by outliers. Best for skewed data.

Mode

- Most frequently occurring value
- Can have 0, 1, or multiple modes
- Only sensible measure for categorical data

MEASURES OF SPREAD

RANGE & IQR

$$\text{Range} = \max - \min$$

$$\text{IQR} = Q_3 - Q_1$$

Q_1 = lower quartile (25%), Q_3 = upper quartile (75%)

STANDARD DEVIATION (Σ)

$$\sigma = \sqrt{[\ \Sigma(x - \bar{x})^2 / n\ ]}$$

Measures spread around mean. Larger σ = more spread.

OUTLIER RULE

$$x < Q_1 - 1.5 \cdot \text{IQR} \text{ OR}$$

$$x > Q_3 + 1.5 \cdot \text{IQR}$$

Points outside these bounds = outliers (box plot rule)

NORMAL DISTRIBUTION

NORMAL DISTRIBUTION $X \sim N(M, \Sigma^2)$

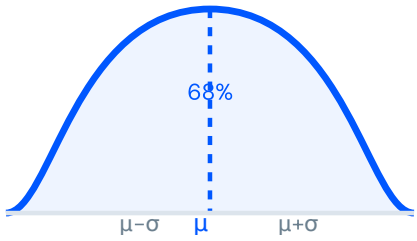
Mean = μ, Standard deviation = σ

68% of data within μ ± 1σ

95% of data within μ ± 2σ

99.7% of data within μ ± 3σ

Bell-shaped, symmetric about mean. Use GDC for all calculations.

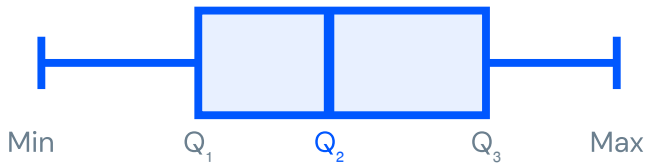


GDC: NORMAL DISTRIBUTION

P(a ≤ X ≤ b): 2nd → DISTR → 2:normalcdf(a, b, μ, σ). **Find x given P:** 2nd → DISTR → 3:invNorm(area, μ, σ). Use -1E99 for -∞, 1E99 for +∞.

Data Representation

BOX & WHISKER PLOT



- Box = IQR (middle 50% of data)
- Line in box = median (Q_2)
- Whiskers extend to min/max (or outlier limits)
- Dots beyond whiskers = outliers

HISTOGRAM



- Area of bar = frequency (not height when unequal widths)
- Modal class = bar with greatest height/area
- Estimate mean using midpoints of each class

CHOOSING THE RIGHT CHART

Chart Type	Best For	Shows
Bar chart	Categorical data	Frequency of categories
Histogram	Continuous data	Distribution shape, spread
Box plot	Comparing distributions	Quartiles, spread, outliers
Scatter diagram	Two variables	Correlation, relationship
Cumulative freq. curve	Finding medians, quartiles	Running totals of frequency
Pie chart	Parts of a whole (categorical)	Proportions

EXAM INSIGHT — DESCRIBING DISTRIBUTIONS

Always comment on: **Shape** (symmetric/skewed), **Centre** (mean or median), **Spread** (range or IQR), **Outliers** (any unusual values). Use context — don't just say "the median is 5".

Correlation, Regression & Hypothesis Testing

CORRELATION

PEARSON'S CORRELATION COEFFICIENT (R)

$$-1 \leq r \leq 1$$

r = 1: perfect positive linear
r = −1: perfect negative linear
r = 0: no linear correlation
|r| > 0.85: strong; 0.5–0.85: moderate; <0.5: weak

REGRESSION LINE (LSRL)

$$\hat{y} = mx + c$$

Always passes through $(\bar{x}, \bar{y})$. Use GDC LinReg(ax+b) for coefficients. Only extrapolate cautiously!

GDC: LINEAR REGRESSION

STAT → EDIT → enter data in L1, L2. Then STAT → CALC → 4:LinReg(ax+b). a = slope, b = y-intercept, r = correlation coeff.

CHI-SQUARED TEST (X^2)

X^2 TEST FOR INDEPENDENCE

$$\chi^2 = \Sigma (O - E)^2 / E$$

O = observed frequency, E = expected frequency
E = (row total × col total) / grand total

Hypothesis Setup

- H_0: Variables are independent (no association)
- H_1: Variables are NOT independent (association exists)
- Degrees of freedom: df = (r−1)(c−1)
- If p-value < significance level → Reject H_0
- Typical significance level: α = 0.05

GDC: X^2 TEST

Enter observed values in matrix [A]. STAT → TESTS → χ^2-Test. GDC calculates χ^2, p-value, and expected values in matrix [B].

WORKED — INTERPRETING REGRESSION

GDC gives $\hat{y} = 2.3x + 15$, r = 0.92. x = years, y = sales (thousands).

1. **Slope 2.3:** For each additional year, sales increase by 2300 on average.
2. **r = 0.92:** Strong positive linear correlation between years and sales.
3. **Predict x=5:** $\hat{y} = 2.3(5)+15 = 26.5$ → approximately 26,500 sales (interpolation = valid)

CORRELATION ≠ CAUSATION

Even with r close to ±1, you cannot conclude one variable CAUSES the other. Always state "there is a strong positive linear correlation between X and Y" — not "X causes Y".

Probability

CORE RULES

BASIC PROBABILITY

$$P(A) = \text{favourable outcomes} / \text{total outcomes}$$

$0 \le P(A) \le 1$. P(certain) = 1. P(impossible) = 0.

COMPLEMENT RULE

$$P(A') = 1 - P(A)$$

"At least one" = 1 − P(none)

ADDITION RULE

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

For mutually exclusive events: $P(A \cap B) = 0$

MULTIPLICATION RULE

$$P(A \cap B) = P(A) \times P(B|A)$$

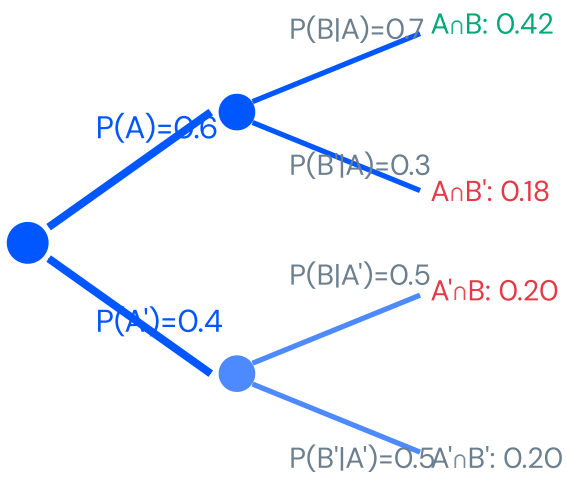
If independent: $P(A \cap B) = P(A) \times P(B)$

CONDITIONAL PROBABILITY

$$P(A|B) = P(A \cap B) / P(B)$$

P(A|B) = probability of A given B has occurred

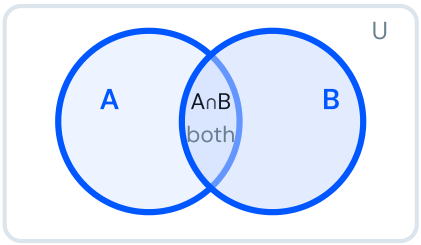
TREE DIAGRAM EXAMPLE



Tree Diagram Rules

- Probabilities on each branch (at each split: must sum to 1)
- End probability = multiply along the path
- P(B) = sum of all paths reaching B
- For conditional P, use the relevant path ÷ total for B

VENN DIAGRAMS



- **A ∩ B** (intersection) = in BOTH circles = AND
- **A ∪ B** (union) = in EITHER circle = OR
- **A only** = P(A) − P(A ∩ B)
- **A' ∩ B'** = neither A nor B = outside both circles
- Sum of all regions = 1 (whole sample space)

Probability Distributions

BINOMIAL DISTRIBUTION

X ~ B(N, P)

$$P(X=r) = C(n,r) \cdot p^r \cdot (1-p)^{n-r}$$

n = number of trials (fixed)
p = probability of success per trial
Each trial independent
E(X) = np, Var(X) = np(1-p)

Conditions for Binomial

- Fixed number of trials (n)
- Only two outcomes per trial (success/fail)
- Constant probability p
- Trials are independent

GDC: BINOMIAL

P(X=r): 2nd → DISTR → binompdf(n,p,r). **P(X≤r):** binomcdf(n,p,r). For P(X≥r): 1 - binomcdf(n,p,r-1).

POISSON DISTRIBUTION HL

X ~ PO(Λ)

$$P(X=r) = e^{-\lambda} \cdot \lambda^r / r!$$

λ = mean rate (events per interval)
E(X) = Var(X) = λ
Use for: rare events over time/space

Conditions for Poisson

- Events occur randomly & independently
- Average rate (λ) is constant
- Events don't occur simultaneously
- Counting events in fixed interval

POISSON APPROXIMATION TO BINOMIAL HL

If n is large, p is small: $\lambda = np$

Rule of thumb: n ≥ 50, p ≤ 0.1

WORKED — BINOMIAL

A fair die is rolled 10 times. Find P(exactly 3 sixes).

1. Identify: X ~ B(10, 1/6), find P(X=3)
2. $P(X=3) = C(10,3) \times (1/6)^3 \times (5/6)^7$
3. $= 120 \times 0.004630 \times 0.2791 \approx 0.155$
4. GDC: binompdf(10, 1/6, 3) = 0.1550

EXAM INSIGHT

"At least 1" = 1 - P(X=0). "At most 3" = P(X≤3) = binomcdf. "More than 3" = 1 - binomcdf(n,p,3). Read carefully!

∂ Calculus: Differentiation

DIFFERENTIATION RULES

POWER RULE

$$\frac{d}{dx}[x^n] = nx^{n-1}$$

d/dx [c] = 0. d/dx [cx] = c.

STANDARD DERIVATIVES

$$\frac{d}{dx}[e^x] = e^x$$

$$\frac{d}{dx}[\ln x] = \frac{1}{x}$$

$$\frac{d}{dx}[\sin x] = \cos x$$

$$\frac{d}{dx}[\cos x] = -\sin x$$

CHAIN RULE HL

$$\frac{d}{dx}[f(g(x))] = f'(g(x)) \times g'(x)$$

e.g. d/dx [(3x+1)⁴] = 4(3x+1)³ × 3 = 12(3x+1)³

PRODUCT & QUOTIENT RULES HL

$$(uv)' = u'v + uv'$$

$$\left(\frac{u}{v}\right)' = \frac{u'v - uv'}{v^2}$$

APPLICATIONS

Gradient at a Point

- Find f'(x), then substitute the x-value
- Gradient of tangent = $f'(x_0)$
- Gradient of normal = $-1/f'(x_0)$

EQUATION OF TANGENT / NORMAL

$$y - y_1 = m(x - x_1)$$

Tangent: m = $f'(x_1)$. Normal: m = $-1/f'(x_1)$

Increasing / Decreasing

- f'(x) > 0 → function is increasing
- f'(x) < 0 → function is decreasing
- f'(x) = 0 → stationary (turning) point

Optimisation Steps

- Form expression for quantity to minimise/maximise
- Differentiate and set f'(x) = 0
- Solve for x; verify with second derivative or table
- f''(x) > 0 → local min; f''(x) < 0 → local max

✎ WORKED — OPTIMISATION

Profit: $P(x) = -2x^2 + 80x - 300$. Find maximum profit and the quantity x.

1. $P'(x) = -4x + 80$
2. Set P'(x) = 0: $-4x + 80 = 0 \rightarrow x = 20$
3. $P''(x) = -4 < 0$ → confirms maximum
4. $P(20) = -2(400) + 80(20) - 300 = -800 + 1600 - 300 = 500$

∫ Calculus: Integration

INTEGRATION RULES

POWER RULE (INDEFINITE)

$$\int x^n \, dx = x^{n+1}/(n+1) + C, \quad n \neq -1$$

Always add constant of integration C

STANDARD INTEGRALS

$$\int e^x \, dx = e^x + C$$

$$\int 1/x \, dx = \ln|x| + C$$

$$\int \cos x \, dx = \sin x + C$$

$$\int \sin x \, dx = -\cos x + C$$

DEFINITE INTEGRAL

$$\int_a^b f(x) \, dx = [F(x)]_a^b = F(b) - F(a)$$

Gives the NET area between curve and x-axis from x=a to x=b

AREA BETWEEN CURVES

$$A = \int_a^b |f(x) - g(x)| \, dx$$

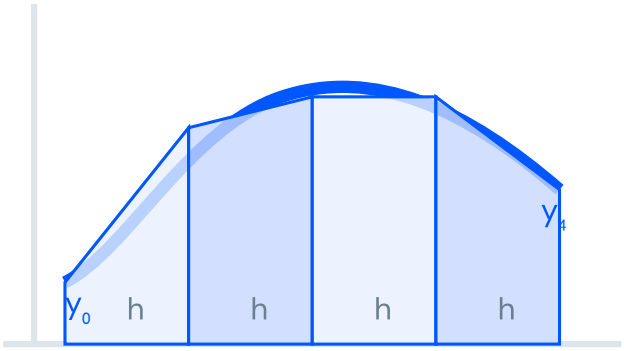
Or: $\int_a^b$ (top - bottom) dx when $f(x) \geq g(x)$

TRAPEZOIDAL RULE (NUMERICAL INTEGRATION)

TRAPEZOIDAL RULE

$$A \approx h/2 \, [y_0 + 2(y_1+y_2+...+y_{n-1}) + y_n]$$

h = (b−a)/n = strip width. y_0, y_n are end values. More strips = more accurate.



APPLICATIONS OF INTEGRATION

Area Under Curve

- Area between curve and x-axis: $\int f(x) \, dx$ from a to b
- If curve goes below x-axis, area is negative — use $\int|$
- Total area = $\int$ above + $\int$ below|

GDC: DEFINITE INTEGRAL

MATH → 9:fnInt(f(X), X, a, b). Or graph function → 2nd → CALC → 7∫f(x)dx, enter lower and upper bounds.

HL Extensions HL ONLY

ADVANCED STATISTICS HL

T-TEST (ONE SAMPLE)

$$t = (\bar{x} - \mu_0) / (s/\sqrt{n})$$

Test if sample mean $\bar{x}$ differs from hypothesised mean μ_0. s = sample SD, n = sample size. Use GDC T-Test.

TWO-SAMPLE T-TEST

$$H_0: \mu_1 = \mu_2 \text{ (no difference in means)}$$

Use GDC 2-SampTTest. Report t-statistic, p-value, and conclusion in context.

SPEARMAN'S RANK CORRELATION HL

$$r_s = 1 - 6\Sigma d_i^2 / (n(n^2-1))$$

d = difference in ranks. Non-parametric. Better for non-linear monotonic data.

DIFFERENTIAL EQUATIONS HL

SEPARABLE DES

$$dy/dx = f(x)g(y) \rightarrow \int 1/g(y)dy = \int f(x)dx$$

Separate variables, integrate both sides, solve for y. Don't forget C.

WORKED — SEPARABLE DE

Solve: dy/dx = 2xy, y(0) = 3

1. $1/y\ dy = 2x\ dx$
2. $\ln|y| = x^2 + C$
3. $y = Ae^{(x^2)}$
4. Apply y(0)=3: $3 = Ae^0 \rightarrow A=3,\ y = 3e^{(x^2)}$

EULER'S METHOD

$$y_{n+1} = y_n + h \cdot f(x_n, y_n)$$

Numerical approximation with step size h. Smaller h = more accurate.

COMPLEX MODELLING HL

LOGISTIC GROWTH

$$L(t) = L_max / (1 + Ae^{(-kt)})$$

L_max = carrying capacity. S-shaped curve. Models limited growth (population).

INTEGRATION BY SUB HL

$$\int f(g(x))g'(x)\ dx$$

Let $u = g(x)$

Reverse of chain rule. Substitute u, integrate in u, substitute back.

MARKOV CHAINS HL

$$s_{n+1} = T \cdot s_n$$

T = transition matrix. Steady state: T·s = s. Solve (T−I)s = 0.

⚠ Common Mistakes to Avoid

1 **Calculator in Degrees Mode**
Always check your GDC is in DEGREE mode for trig (unless radians specified). MODE → check 3rd line shows "Degree".

2 **Forgetting the Constant of Integration**
Indefinite integrals MUST include "+ C". You lose marks without it. Always write C.

3 **Confusing Correlation with Causation**
r close to ±1 shows strong correlation, NOT that one variable causes the other. Use "linear correlation" language.

4 **Wrong Direction for Graph Transformations**
f(x−3) shifts RIGHT by 3, NOT left. f(x)+3 shifts UP by 3. The horizontal shift is counterintuitive.

5 **Misidentifying GP vs AP**
AP: constant DIFFERENCE between consecutive terms. GP: constant RATIO. Always test: subtract for AP, divide for GP.

6 **Using Regression Line Outside Data Range**
Extrapolation beyond the data range is unreliable. Must comment on this limitation when extrapolating.

7 **Ambiguous Case in Sine Rule (SSA)**
If given two sides and non-included angle, there may be TWO possible triangles. Check if angle > 90° or if other angle possible.

8 **Not Showing Method (Working)**
Even if using GDC, write the formula/setup. "Checked on GDC" alone scores zero. Show each step.

9 **Rounding Too Early**
Keep full precision in intermediate steps. Only round the FINAL answer (usually 3 significant figures unless stated).

10 **Area Below x-axis = Negative Integral**
Definite integral gives negative value if curve is below x-axis. For AREA, take absolute value: $|\int f(x)\,dx|$.

11 **Missing P(X≥r) vs P(X>r) Distinction**
P(X≥3) = 1 − P(X≤2). P(X>3) = 1 − P(X≤3). These are DIFFERENT. Read "at least" vs "more than" carefully.

12 **Forgetting to Check Convergence of Infinite GP**
$S_\infty = u_1/(1-r)$ ONLY valid when $|r| < 1$. If $|r| \geq 1$, the series diverges (no sum).

13 **Wrong df in Chi-Squared Test**
df = (rows−1)(columns−1). Don't include totals row/column in count. df is NOT the number of cells.

14 **Differentiating When Should Integrate (and Vice Versa)**
Rate of change problems: differentiate. Area/accumulation problems: integrate. Velocity→displacement: integrate.

🏆 Exam Strategy

PAPER STRUCTURE

Paper	SL	HL	Calc?
Paper 1	80 min, 80 marks	120 min, 110 marks	❌ No
Paper 2	80 min, 80 marks	120 min, 110 marks	✅ Yes
Paper 3	—	60 min, 55 marks	✅ Yes

TIME MANAGEMENT

Per-Mark Rule

- Aim for ~1 minute per mark
- 5-mark question → <5 minutes
- If stuck after 2 minutes, move on and return
- Last 5 minutes: check all answers
- Never leave a question blank — partial marks!

PAPER 1 STRATEGY (NO CALC)

- Show ALL algebraic working — full marks depend on method
- Use the formula booklet — it's provided!
- Exact answers preferred: leave in surd form or as fractions
- Sketch graphs: label key points (intercepts, vertex)
- Check reasonableness: does your answer make sense?

GDC STRATEGY (PAPER 2)

🔢 **BEFORE THE EXAM**

Check batteries. Confirm DEGREE mode. Clear RAM only if needed. Know locations of: STAT, DISTR, CALC, MATH, APPS→Finance.

Essential GDC Skills

- **Graph:** Enter Y=, set WINDOW, press GRAPH
- **Zeros:** 2nd+CALC → 2:zero, move left/right of root
- **Intersect:** 2nd+CALC → 5:intersect
- **LinReg:** STAT→CALC→4:LinReg(ax+b)
- **Normalcdf:** 2nd→DISTR→2:normalcdf(l, u, μ, σ)
- **TVM:** APPS→Finance→TVM Solver
- **χ^2:** STAT→TESTS→χ^2-Test, matrix [A]
- **Integral:** MATH→9:fnInt or graph 2nd+CALC→7

ANSWER WRITING TIPS

💡 **CONTEXT IS KEY**

IB questions often ask to "interpret" results in context. Always tie your answer back to the real-world scenario. Never just write a number — say what it means.

- Start each question on a new line
- Label diagrams clearly with values and units
- State any assumptions you make
- Define variables when setting up models
- Write conclusion sentences for hypothesis tests

⚡ Top Formulas Quick Reference

NUMBER & ALGEBRA

AP: NTH TERM
$u_n = u_1 + (n-1)d$

AP: SUM
$S_n = n/2(2u_1 + (n-1)d)$

GP: NTH TERM
$u_n = u_1 \cdot r^{n-1}$

GP: INFINITE SUM
$S_\infty = u_1/(1-r),\ |r|<1$

COMPOUND INTEREST
$A = P(1 + r/n)^{(nt)}$

DEPRECIATION
$V = P(1 - r)^t$

GEOMETRY & TRIG

SINE RULE
$a/\sin A = b/\sin B = c/\sin C$

COSINE RULE
$a^2 = b^2+c^2-2bc\cdot\cos A$

TRIANGLE AREA
$A = \frac{1}{2}ab\cdot\sin C$

CYLINDER VOLUME
$V = \pi r^2 h$

SPHERE VOLUME
$V = 4/3\ \pi r^3$

CONE VOLUME
$V = 1/3\ \pi r^2 h$

STATS, PROB & CALCULUS

CONDITIONAL PROBABILITY
$P(A|B) = P(A\cap B)/P(B)$

BINOMIAL
$P(X=r) = C(n,r)p^r(1-p)^{n-r}$

NORMAL: E(X)
$X \sim N(\mu, \sigma^2)$

CHI-SQUARED
$\chi^2 = \Sigma(O-E)^2/E$

POWER RULE (DIFF)
$d/dx[x^n] = nx^{n-1}$

POWER RULE (INT)
$\int x^n dx = x^{n+1}/(n+1)+C$

QUADRATIC FORMULA & KEY RELATIONSHIPS

$$x = (-b \pm \sqrt{(b^2-4ac)}) / 2a$$

$$\int_a^b f(x)dx = F(b) - F(a)$$

$$A = P(1 + r/n)^{(nt)}$$

💡 **FORMULA BOOKLET REMINDER**

The IB provides a formula booklet in the exam. Know WHAT formulas are in it so you don't waste time re-deriving them. Also know WHAT ISN'T in it (e.g. Pearson's r formula, percentage change, AP sum formula — verify your edition).

🌙 Night Before Exam — Final Revision

TOPIC SUMMARY BULLETS

📐 Number & Algebra

AP: add d. GP: multiply r. Compound: A=P(1+r/n)^nt. Depreciation: V=P(1−r)^t

📈 Functions

Quadratic: roots via formula; vertex x=−b/2a. Exponential: b>1 growth, b<1 decay. Transformations: f(x−h) shifts RIGHT.

📐 Geometry

SOH CAH TOA. Sine rule: a/sinA. Cosine rule for SAS/SSS. Area=½ab·sinC. Volumes in formula booklet.

📊 Statistics

Normal: 68-95-99.7. r close to ±1 = strong. LSRL passes through $(\bar{x},\bar{y})$. χ^2: reject H_0 if $p < \alpha$.

🎲 Probability

Complement: P(A')=1−P(A). Tree: multiply along branches, add paths to B. Binomial: B(n,p). "At least 1" = 1−P(0).

∫ Calculus

Differentiate: nx^{n-1}. Integrate: $x^{n+1}/(n+1)+C$. Max/min: set f'=0, check f''. Definite integral = area.

NIGHT BEFORE CHECKLIST

- ☐ Calculator has fresh batteries and is in DEGREE mode
- ☐ Formula booklet located and checked
- ☐ Revision of your 3 weakest topics done
- ☐ GDC skills practiced: LinReg, normalcdf, TVM
- ☐ Sleep 8 hours — recall improves significantly
- ☐ Stationery ready: pencils, ruler, compass
- ☐ Alarm set — arrive 15 minutes early

MORNING OF EXAM

- ☐ Eat a proper breakfast — sustained energy
- ☐ Skim this page only — no heavy review
- ☐ Check GDC mode one last time before entering exam
- ☐ Read each question TWICE before answering
- ☐ Look at marks available — match effort accordingly
- ☐ Don't leave any question blank
- ☐ Review your answers in last 5 minutes

⭐ **FINAL MINDSET**

You've prepared. Trust your work. For each question: **Read → Identify type → Recall method → Execute → Check**. Partial marks are real — attempt everything.

Good luck! 🎯

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